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Poisson Regression Basics

Poisson Regression Inference and Goodness of Fit

- **Video:** Goodness of Fit for Poisson Regression I
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  1h
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  Submitted

Assignments: GLMs for Count Data

Poisson Regression Inference and Goodness of Fit

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Due Jun 2, 11:59 PM IST

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1. The "deviance" of a Poisson regression model is -2 times the log likelihood of the Poisson regression model evaluated at the maximum likelihood estimates. **3 / 3 points**

- (•) True
- ( ) False

Correct

2. The null deviance is the deviance for the model with just an intercept term. **3 / 3 points**

- (•) True
- ( ) False

Correct

3. The saturated model is the model where each data point has it's own unique parameter. **3 / 3 points**

- (•) True
- ( ) False

Correct

4. The residual deviance can be used to test the hypotheses **3 / 3 points**

$$H_0 : \text{The model with } p \text{ predictors fits well enough.} \quad vs.$$

$$H_1 : \text{The model with } p \text{ predictors does not fit well enough.}$$

- (•) True
- ( ) False

Correct

5. A plot of the deviance residuals against the linear predictor (η_i) can provide evidence of a lack of fit of a Poisson regression model. **3 / 3 points**

- (•) True
- ( ) False

Correct

6. Consider a model that attempts to explain the number of awards earned by students at a high school in a year based on their math final exam score and the type of program that they are enrolled in. The categorical predictor variable has three levels indicating the type of program in which the students is enrolled. The categorical predictor levels are "Remedial", "Standard" and "Honors". Here's some output from a Poisson regression. **4 / 4 points**

Consider fitting two models, one with both predictors, and one with just math final exam score as a predictor.

Model 1: num_awards ~ math
Model 2: num_awards ~ prog + math

Resid. Df	Resid. Dev	Df	Deviance	Pr(>Chi)
198	204			
196	189	2	14.6	0.00069 ***

- [ ] The conclusion of this test is that the program variable is not statisticallly significant.
- [ ] The hypotheses under consideration are:

  $$H_0 : \text{The model with math score and program fits well enough.} \quad vs.$$

  $$H_1 : \text{The model with just math score fits well enough.}$$

- [x] The hypotheses under consideration are:

  $$H_0 : \text{The model with just math score fits well enough.} \quad vs.$$

  $$H_1 : \text{The model with just math score does not fit well enough.}$$

  Correct

- [x] The conclusion of this test is that the program variable is statisticallly significant.

  Correct

- [x] The test performed was a χ^2 test.

  Correct

7. Consider a Poisson regression model with the response of the total number of cyclist counts at Manhattan Bridge in a 24 hour period. But suppose that cyclist counts on this bridge are such that, if an individual cycles over the Manhattan Bridge on a particular day, that individual will be *more likely* to cycle over the Manhattan Bridge the following day. So, an event occuring on one day impacts the probability of the event occuring on the next day. The distribution of the number of cyclists over the Manhattan Bridge will then be overdispersed with respect to the Poisson model. **3 / 3 points**

- (•) True
- ( ) False

Correct

8. Which of the following are potential causes of overdispersion? **4 / 4 points**

- [ ] A non-normal predictor variable.
- [x] A dependent response variable.

  Correct

- [x] Outliers.

  Correct

- [x] A missing predictor variable.

  Correct

- [x] Having many zeros recorded for the response.

  Correct